Solving quadratic equations using sq[redacted] roots

1 Determine whether the given value of x is a solution to the given quadratic equation.

a $5x^2+4x=-12\ ;\ x=-2$

b $48x^2+16x-7=0\ ;\ x=\dfrac{1}{4}$

c $(7x-2)^2=37\ ;\ x=\dfrac{8}{7}$

d $20x^2=3-11x\ ;\ x=-\dfrac{3}{4}$

2 If $u^2=d$, what does the square root property state that u is equal to?

3 The equation $x^2-144=0$ has a positive integer solution $x=12$.
Find the negative integer that is also a solution.

4 Find the value of the variable for each of the following equation:

a $x^2=25$

b $x^2=81$

c $x^2=98$

d $x^2=11$

e $3y^2=12$

f $25y^2=36$

g $x^2=10$

h $-3k^2=-12$

i $x^2-25=0$

j $x^2-5=31$

k $\dfrac{x^2}{25}-3=6$

l $81x^2-64=0$

m $-25v^2+36=0$

n $5\left(p^2-3\right)=705$

o $(x-7)^2=81$

p $(x+7)^2=121$

q $(4-x)^2=9$

r $(x-4)^2=10$

s $(4x+3)^2=64$

t $5x^2-15=0$

5 How many real solutions does the equation $x^2+64=0$ have?

6 A quadratic equation has solutions $x=\pm5$.
What must the equation be?

7 Solve $ax^2-c=0$ for x.
Assume a and c are positive.

8 Solve for k:

$$c_1^2k^2+4=40$$

9 On the graph of $y=x^2-4$, there are two points where $y=12$.
Find the x coordinates of these two points.

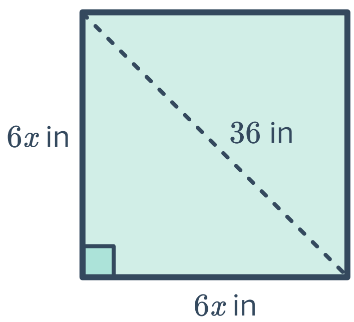
10 The Widget and Trinket Emporium has relea[illegible]he forecast of its revenue for the next year. The revenue R (in dollars) at any point in time t (in months) is described by the equation $R = -(t-18)^2 + 16$.

Find the times at which the revenue will be zero.

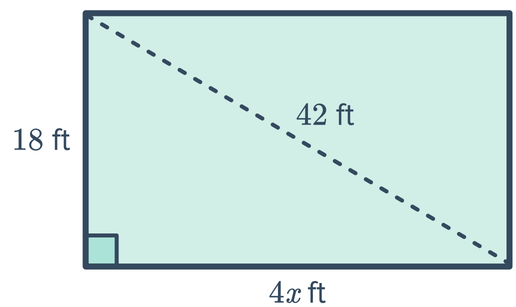
Additional questions

11 Solve for x:

a



b



12 On Earth, the equation $d = 4.9t^2$ is used to find the distance (in meters) an object has fallen through the air after t seconds.

Willow is sky diving and wants to release her paraachute once she has fallen 400 m

Determine the time it will take her to fall 400 m, rounding your answer to the nearest second.

13 The Panorama Tower in Miami, Florida, is 86[redacted]ll.
We can model the behaviour of objects falling from the arch using Galileo's formula for falling objects: $d = 16t^2$, where d is distance fallen in feet and t is time in seconds since the object was dropped.
How long would it take an object dropped from the arch to fall to the ground? Round your answer to the nearest tenth.

14 The kinetic energy E of a moving object is given by $E = \frac{1}{2}mv^2$, where m is its mass in kilograms and v is its speed in meters/second.
If a vehicle weighing 1600 kilograms has kinetic energy $E = 204\,800$, at what speed is it moving?

15 Eduardo is trying to solve the equation $(x+7)^2 = 28$. He thinks that the equation is equivalent to $x + 7 = 2\sqrt{7}$. Nicolette, however, thinks he has performed the operations incorrectly, and that he should have subtracted 7 from both sides of the equation first giving an equivalent equation of $x^2 = 21$
Describe any errors Eduardo and/or Nicolette have made, and find the correct solution to the equation.

16 Consider the quadratic equation $ax^2 - c = 0$ for. Given the following conditions, state the real solution(s). If there are no real solutions state "No real solutions"

a a and c are positive.

b a and c are negative.

c a and c have opposite signs.

d c is equal to 0.